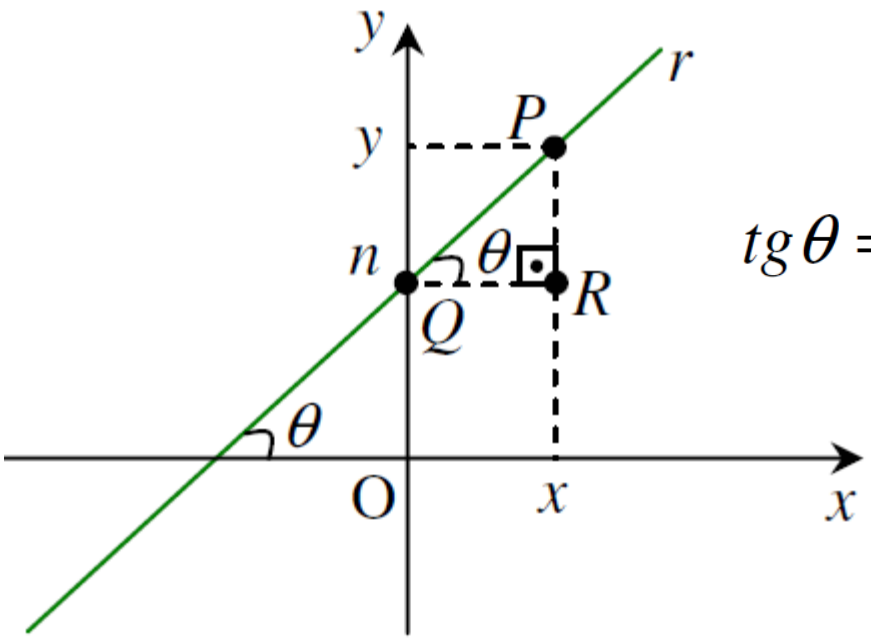
Retas, Planos , Cônicas e Quádricas

(uma breve revisão)

Retas no plano

$$ax + by + c = 0$$

$$y = mx + n$$ Equação reduzida



$$tg\,\theta = m$$

$$y - y_0 = m(x - x_0)$$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}$$

Retas no espaço

Equação simétrica

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

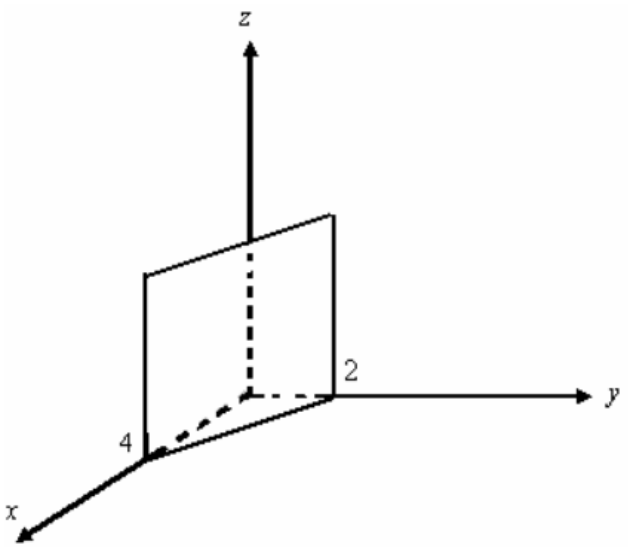
Equação reduzida

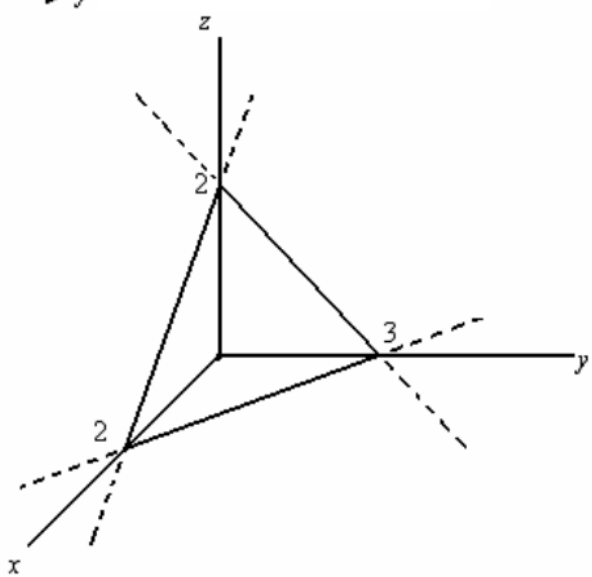
$$\begin{cases} y = m_1 x + n_1 \\ z = p_1 x + q_1 \end{cases}$$

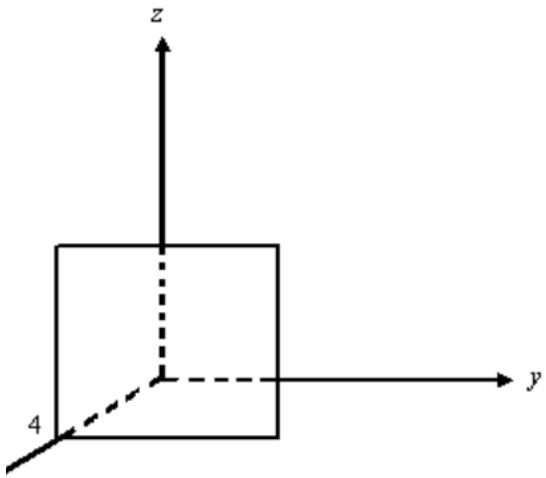
Planos

Equação geral do plano

$$ax + by + cz + d = 0$$







- Qual a representação geométrica da equação *X=2* considerando:

$$R$$

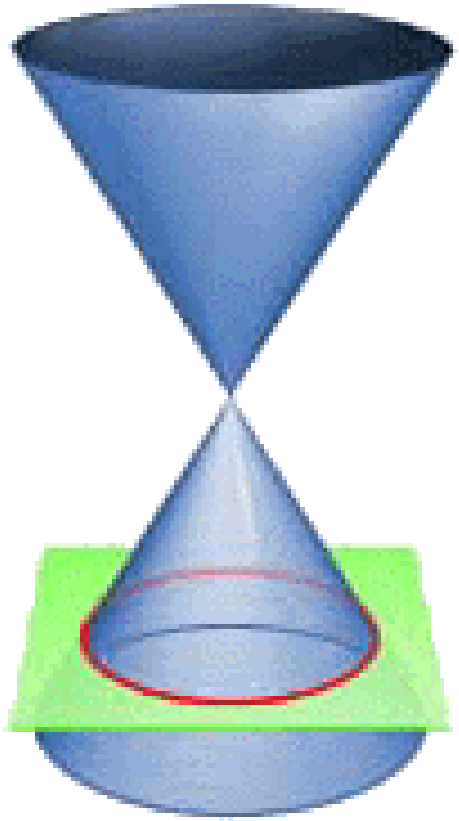
$$R^2$$

$$R^3$$

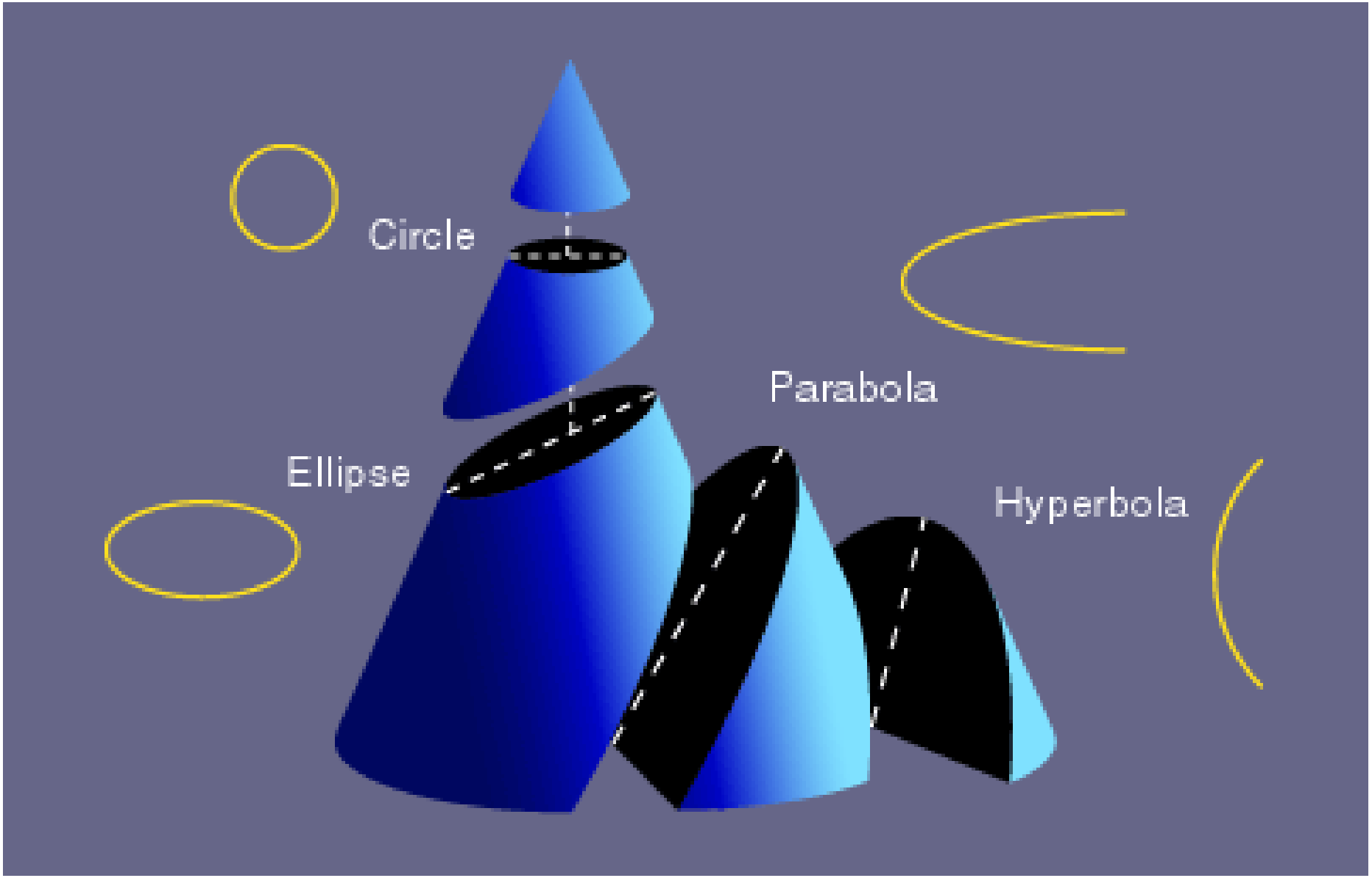
$$R^4$$

$$\vdots$$

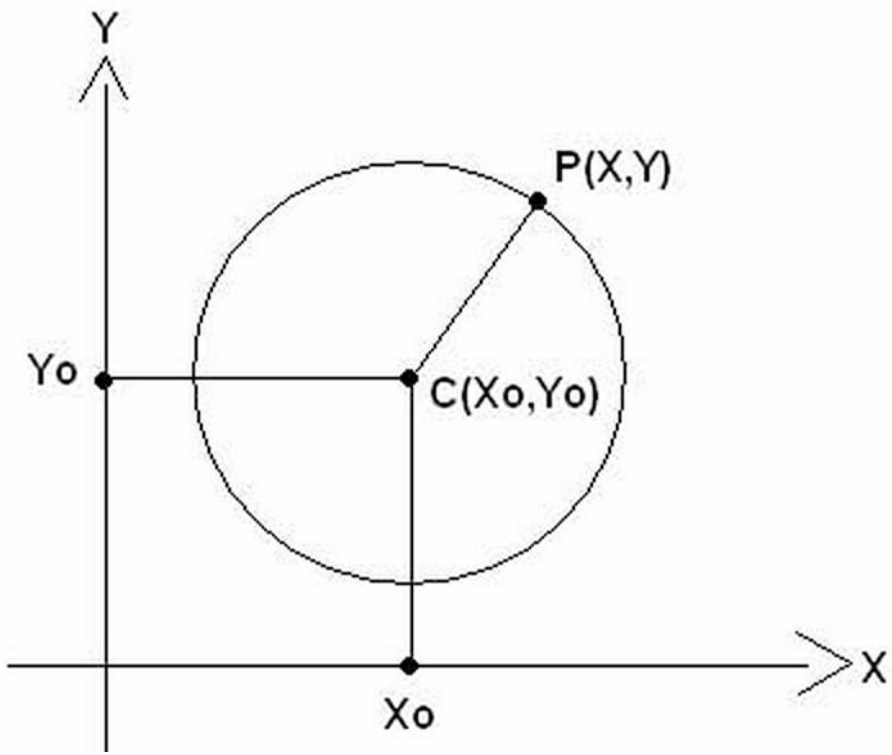
Seções Cônicas



Circle


Circle
Parabola
Ellipse
Hyperbola

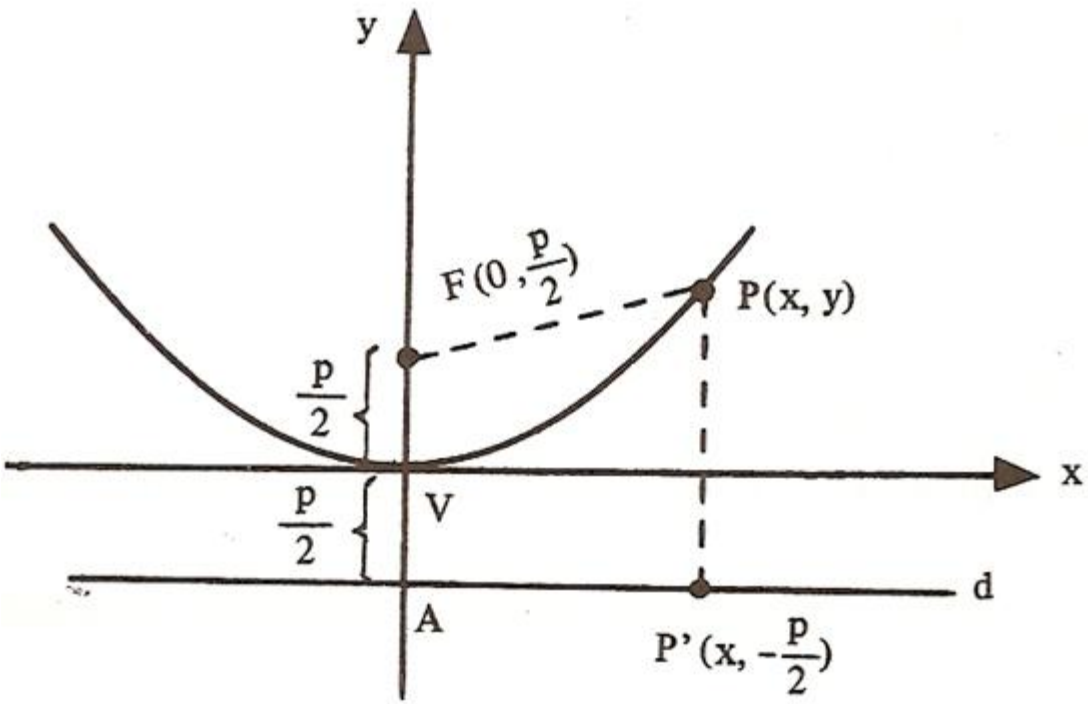
Circunferência



$$(x_1 - x_0)^2 + (y_1 - y_0)^2 = r^2$$

Circunferência com centro em (x_0, y_0) e raio r.

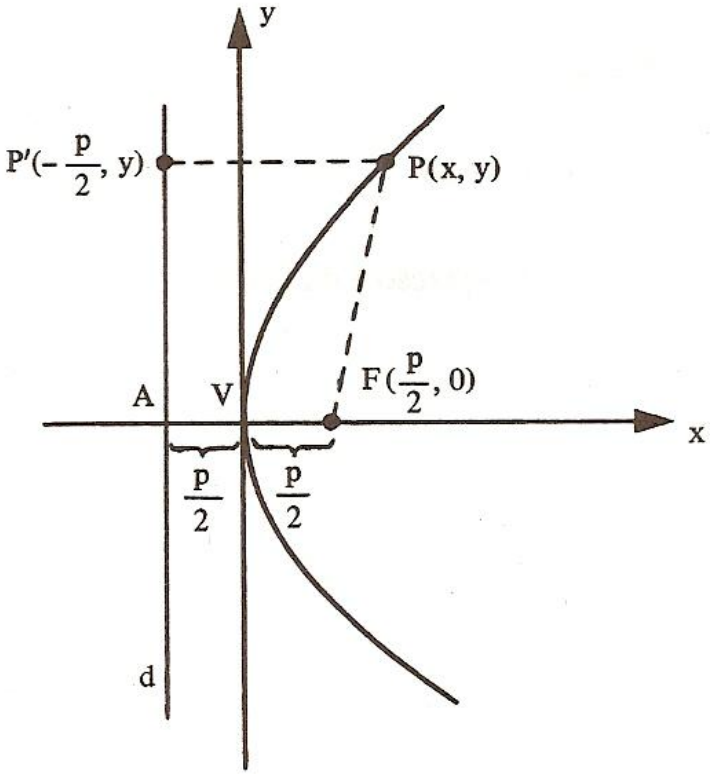
Parábola



$$x^2 = 2py$$

$$ax^2 + bx + c = 0$$

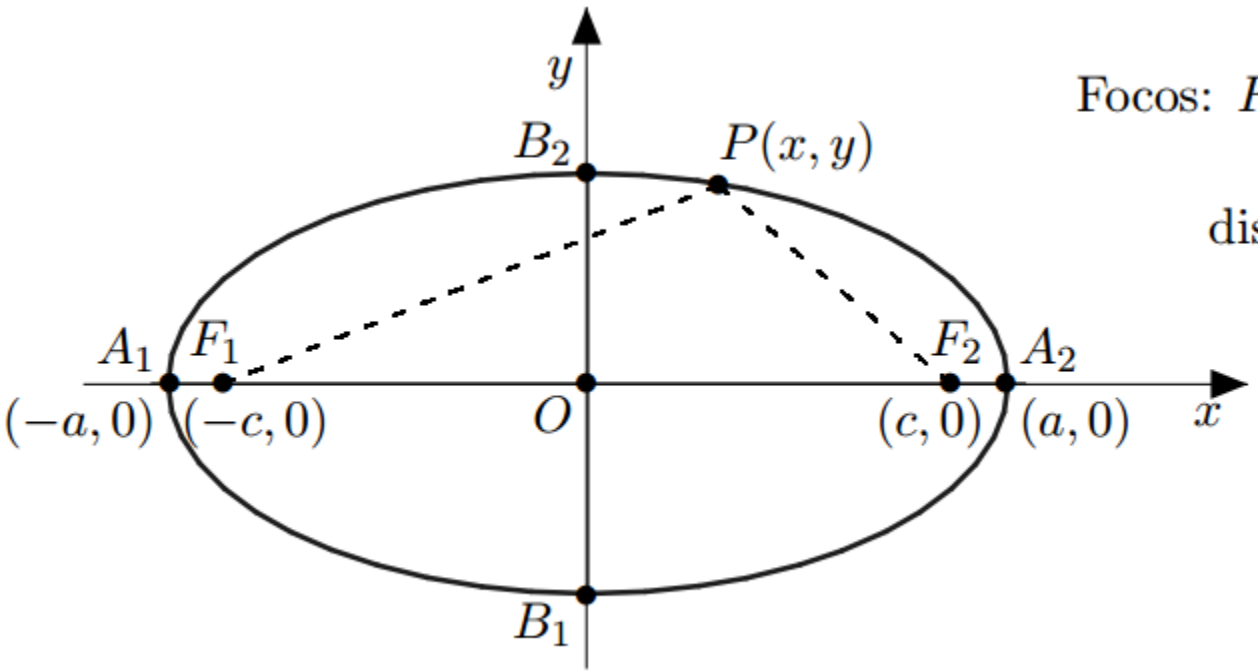
Parábola



$$y^2 = 2px$$

$$ay^2 + by + c = 0$$

Elipse



Focos: $F_1 = (-c, 0)$ e $F_2 = (c, 0)$

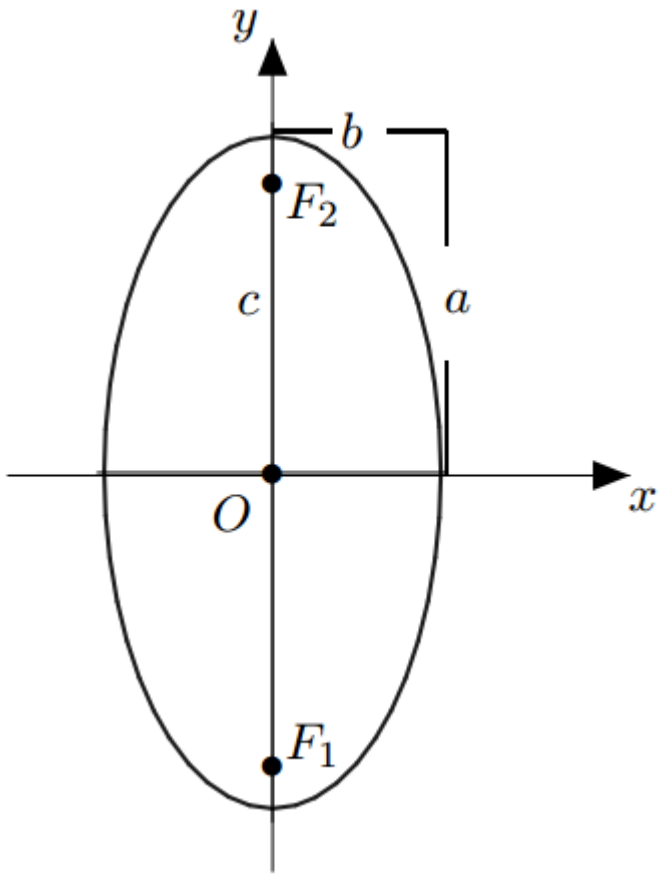
$$\text{dist}(P, F_1) + \text{dist}(P, F_2) = 2a$$

Ox: eixo focal

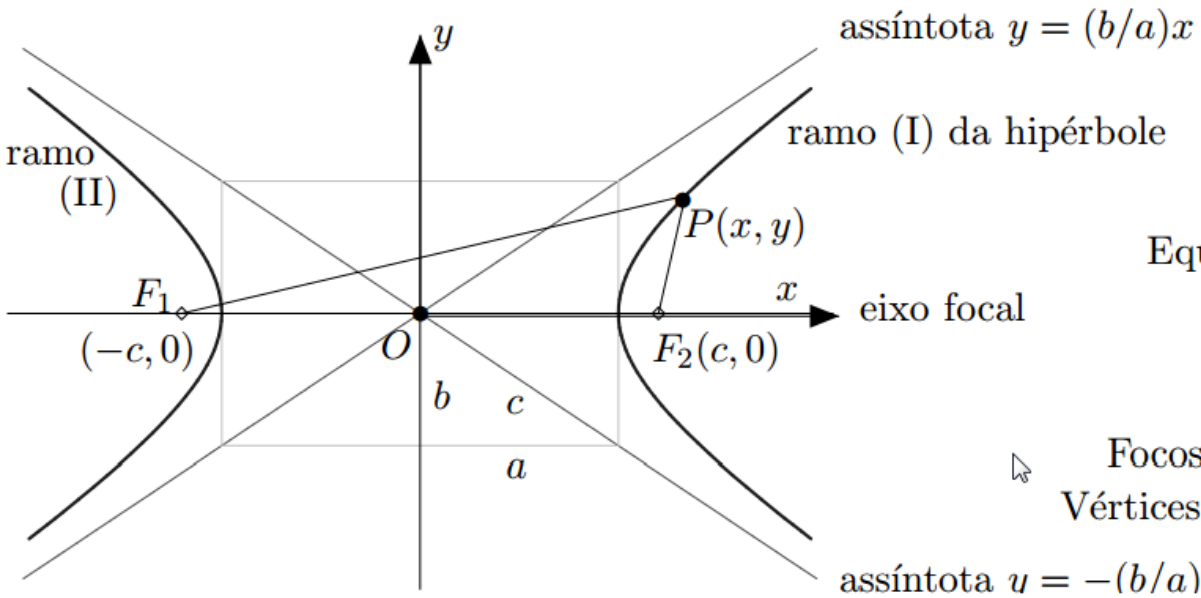
Equação $$\frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} = 1$$

Elipse

Equação $$\frac{(x-x_0)^2}{b^2}+\frac{(y-y_0)^2}{a^2}=1 \quad \text{com } a>b \text{ e } a^2-c^2=b^2$$



Hipérbole



Equação: $\boxed{\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1}$

$b^2 = c^2 - a^2$

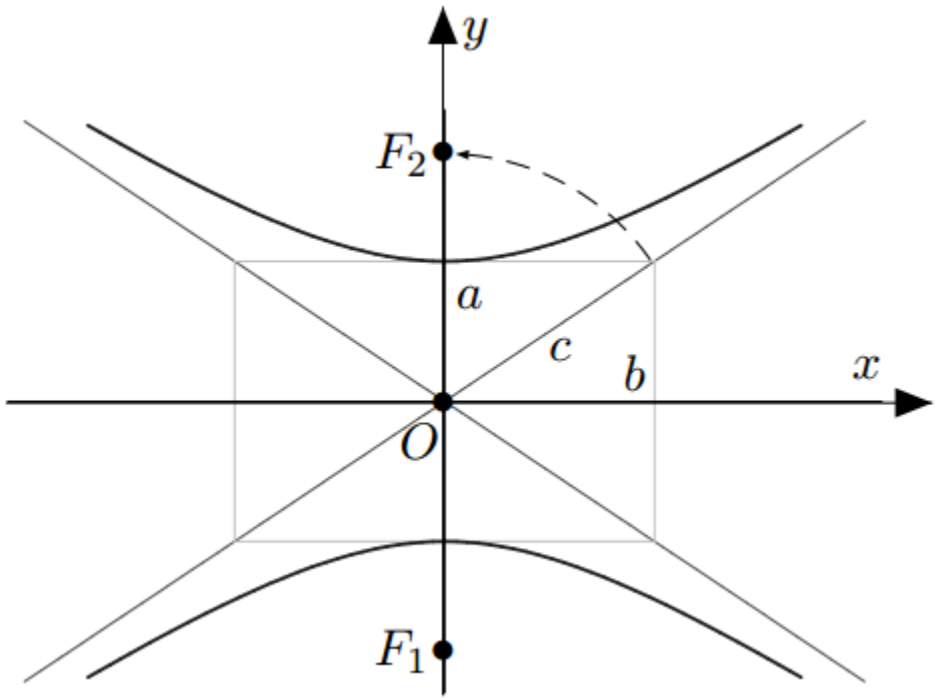
Centro: $O = (0, 0)$

Focos: $F_1(-c, 0)$ e $F_2(c, 0)$

Vértices: $V_1(-a, 0)$ e $V_2(a, 0)$

$$\frac{(x - x_0)^2}{a^2} - \frac{(y - y_0)^2}{b^2} = 1$$

Hipérbole



Hipérbole $\boxed{-\dfrac{x^2}{b^2}+\dfrac{y^2}{a^2}=1}$

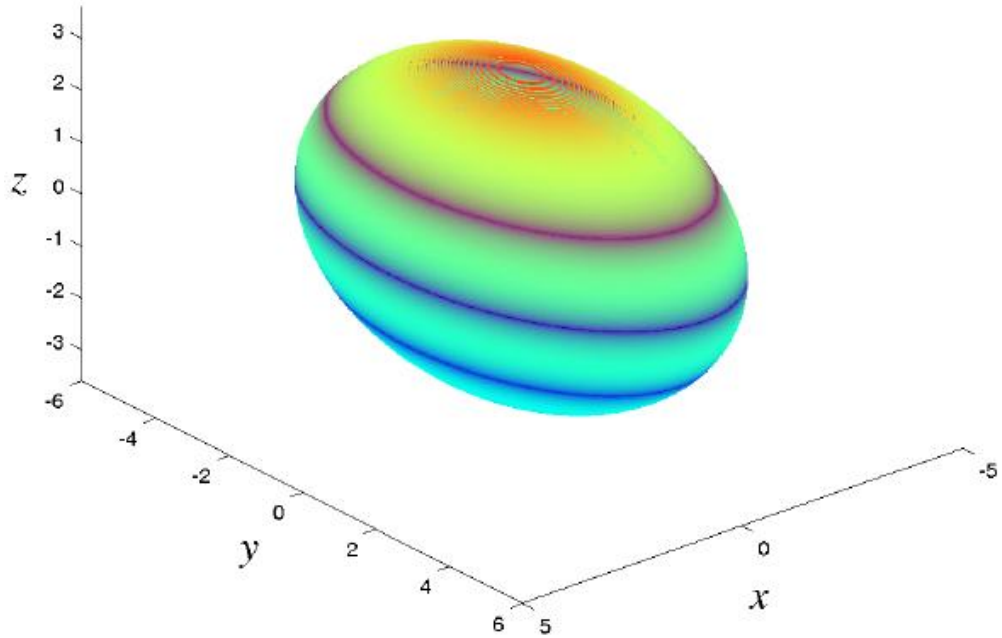
$b^2 = c^2 - a^2$

O = centro = ponto de simetria
Oy = eixo focal
Focos: $F_1(0,-c)$ e $F_2(0,c)$

$$-\frac{(x-x_0)^2}{b^2}+\frac{(y-y_0)^2}{a^2}=1$$

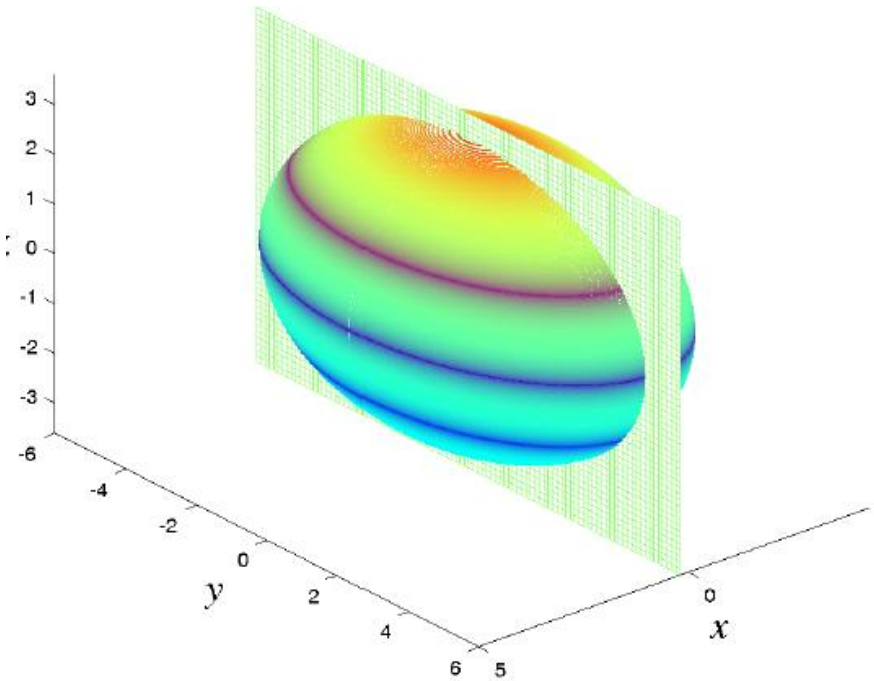
Quádricas

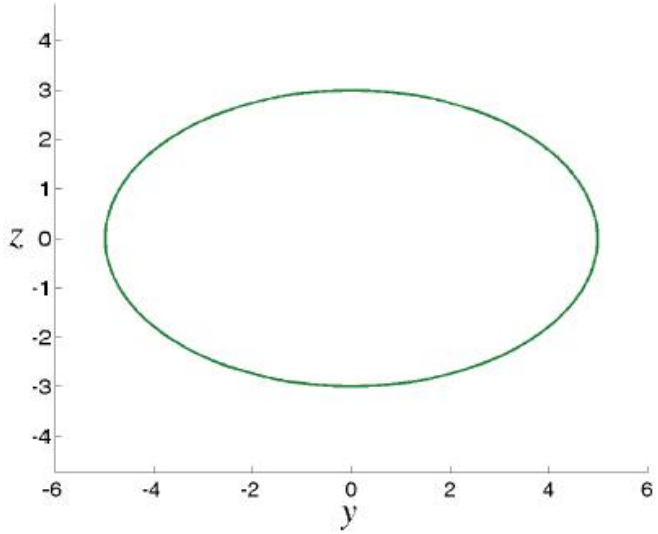
Elipsóide



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Elipsóide





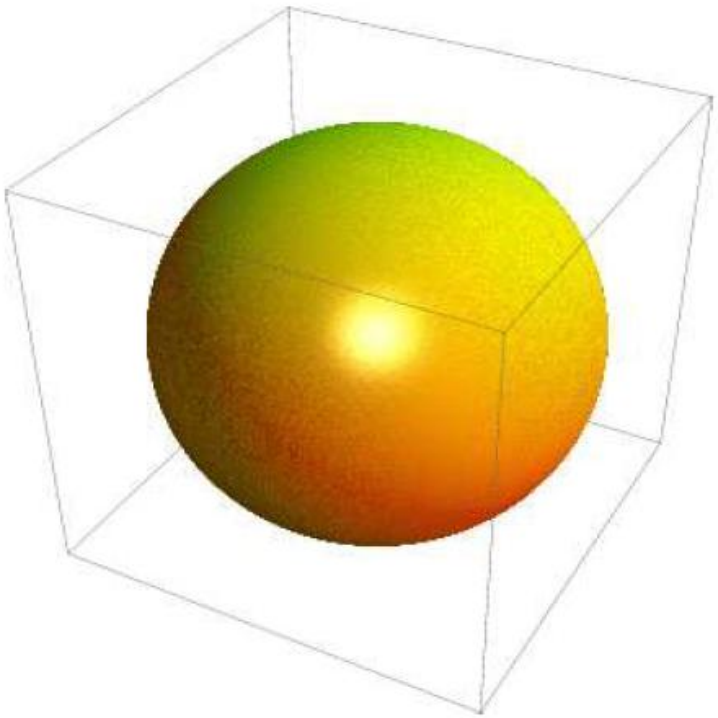
$$\begin{cases} \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1 - \dfrac{k^2}{a^2} \\ x = k \end{cases} \quad : \quad \text{elipses } (-a < k < a).$$

Esfera : caso particular do elipsóide

Em particular se $a = b = c$, temos:

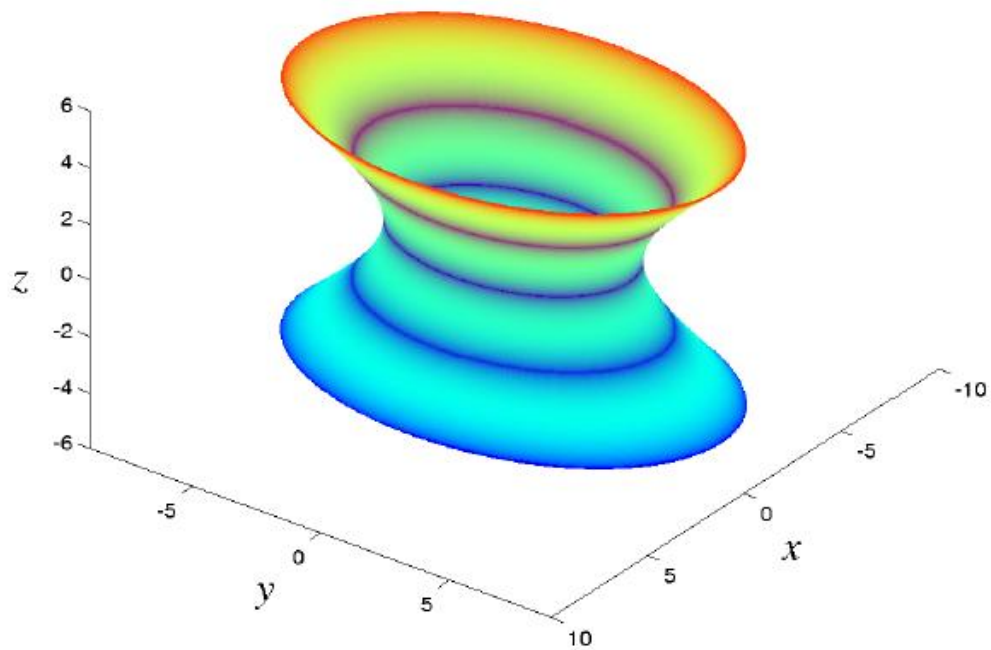
$$\boxed{x^2 + y^2 + z^2 = a^2}$$

equação que representa a esfera de centro na origem e raio a.



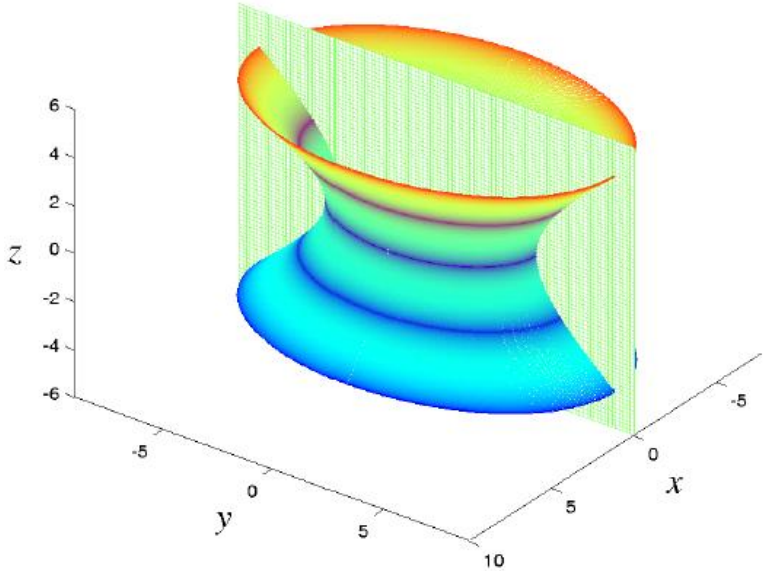
$$\boxed{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2}$$

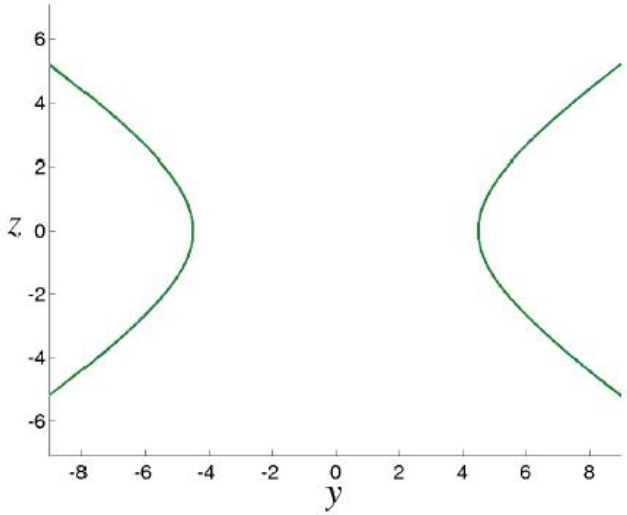
Hiperbolóide de uma folha



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

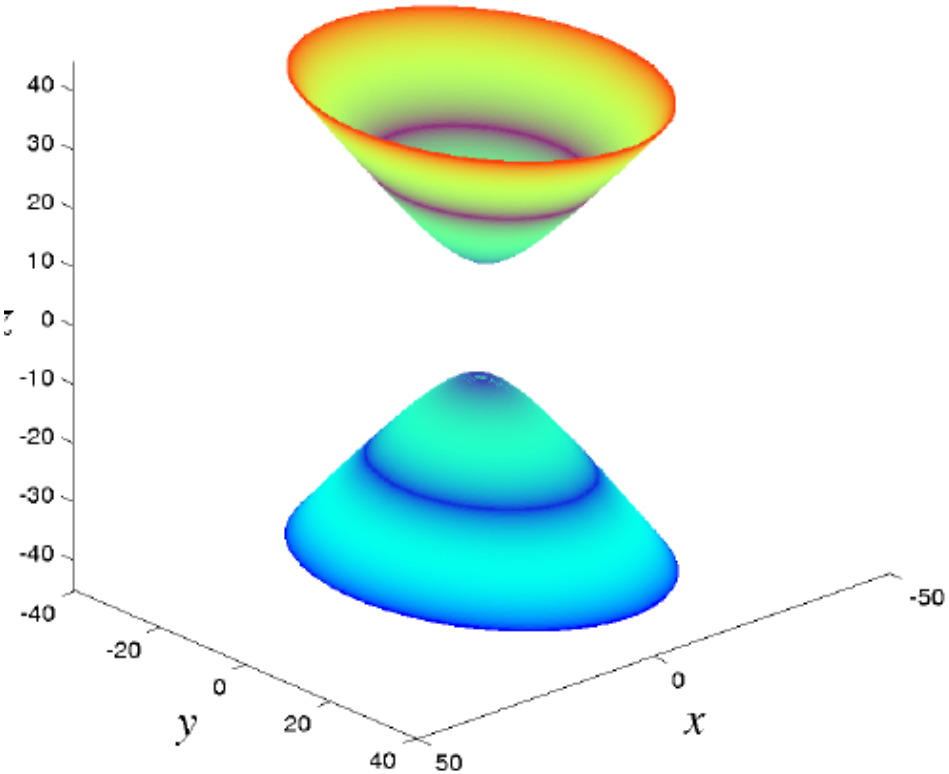
Hiperbolóide de 1 folha





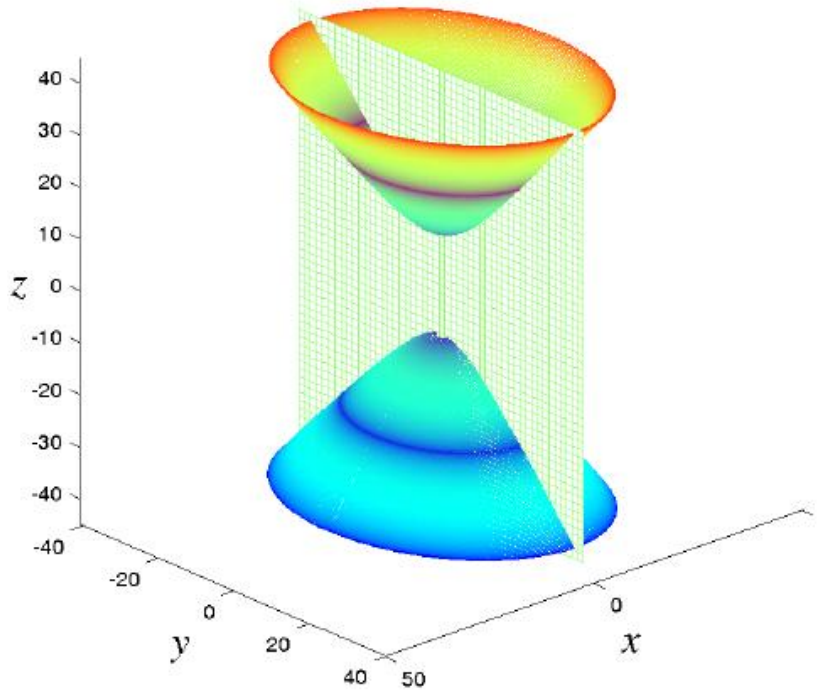
$$\begin{cases} \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1 - \dfrac{k^2}{a^2} \\ x = k \end{cases} \quad : \quad \text{hipérboles } (-a < k < a).$$

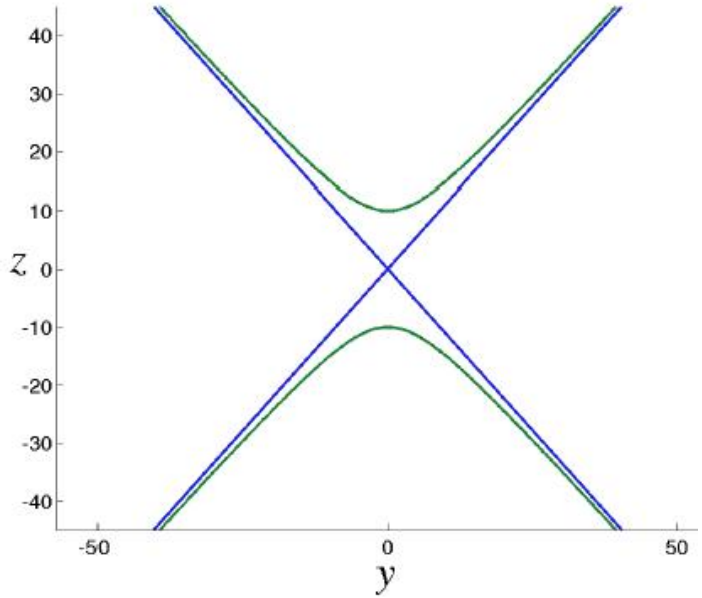
Hiperbolóide de 2 folhas



$$-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

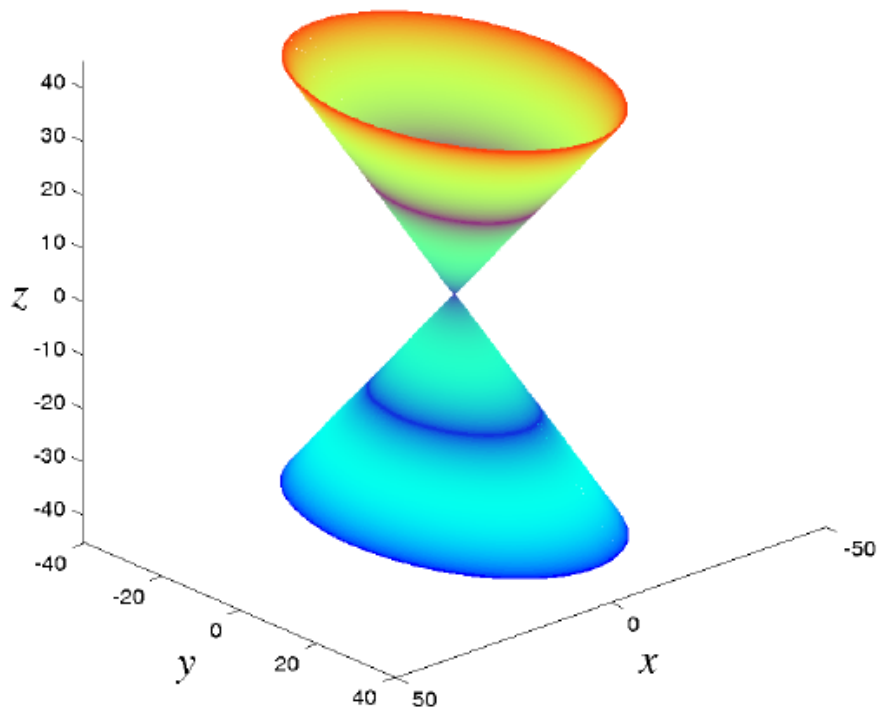
Hiperbolóide de 2 folhas





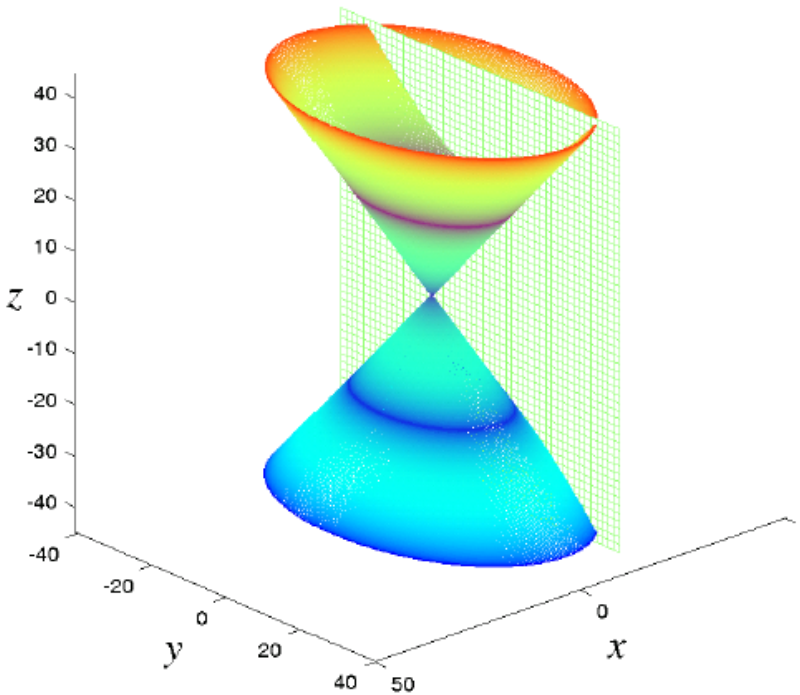
$$\begin{cases} -\dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1 + \dfrac{k^2}{a^2} \\ x = k \end{cases} \quad : \quad \text{hipérboles } \forall k \in \mathbb{R}.$$

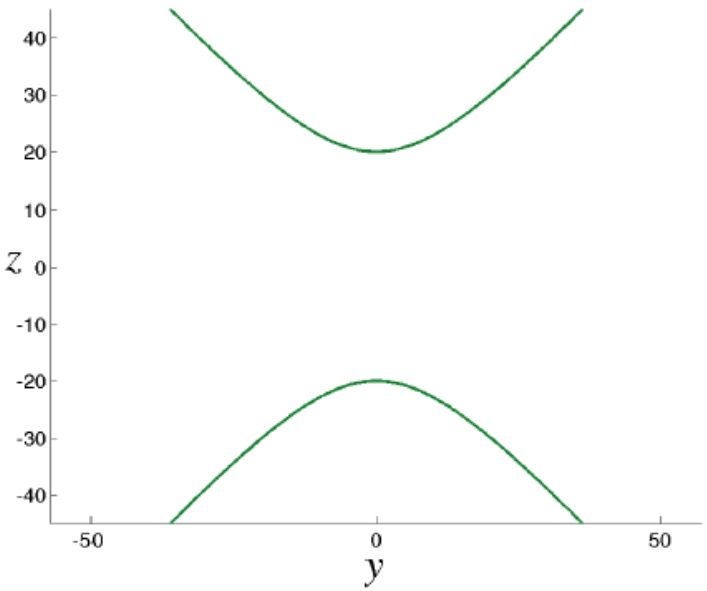
Cone



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

Cone





$$\begin{cases} -\dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = \dfrac{k^2}{a^2} \\ x = k \end{cases} \quad : \quad \text{hipérboles } \forall k \neq 0.$$

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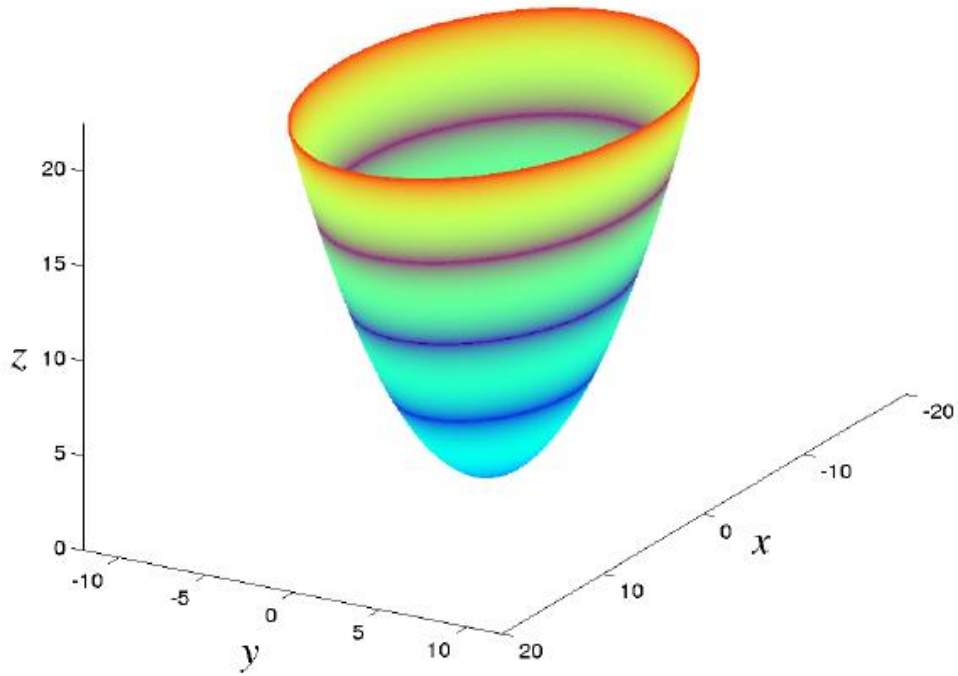
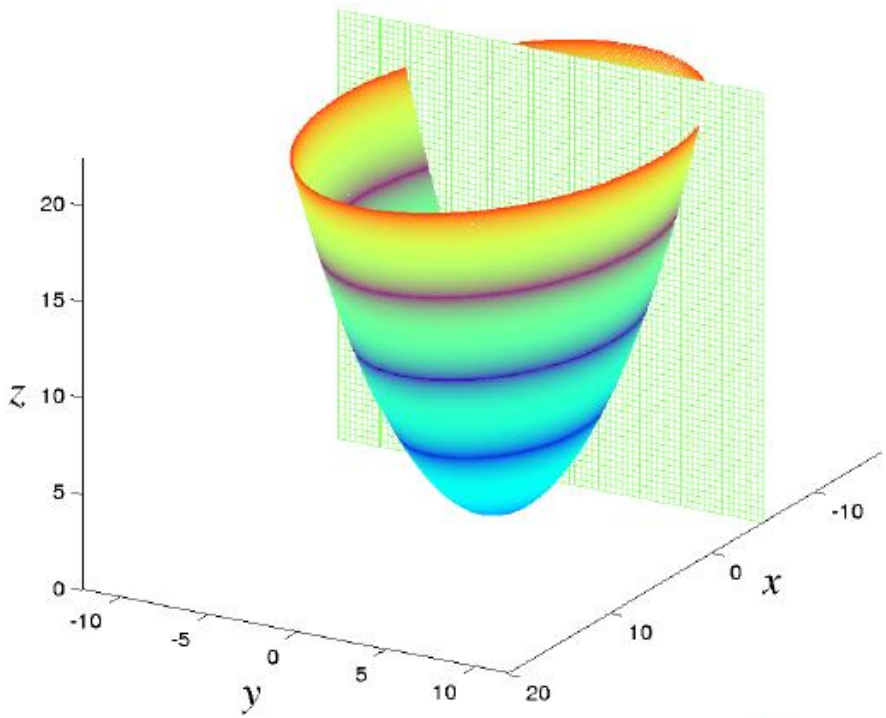
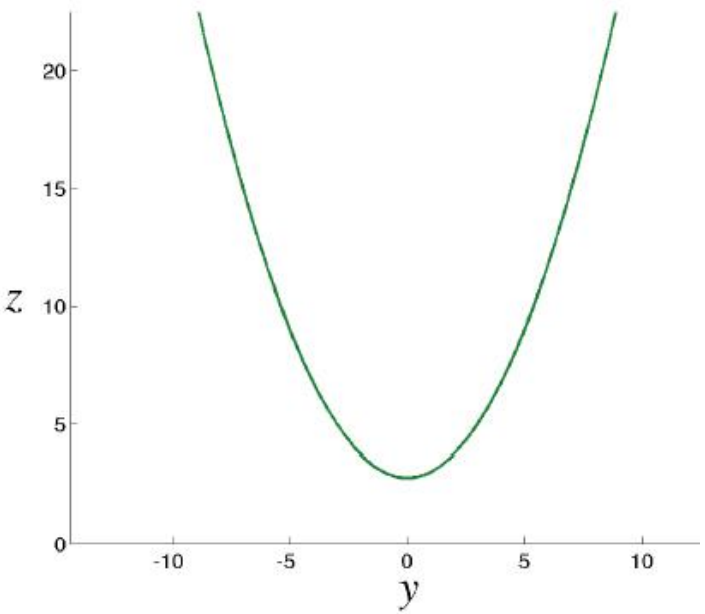
Parabolóide Elíptico



$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

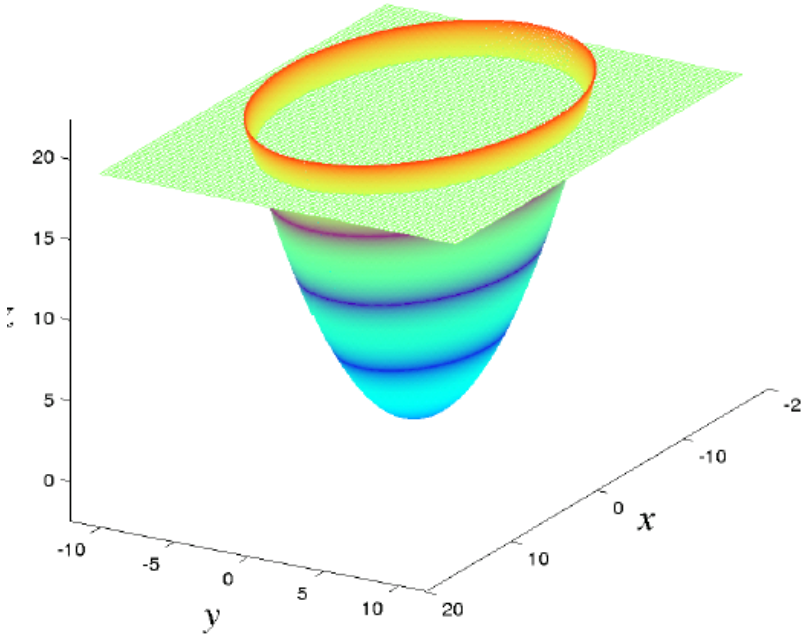
Parabolóide Elíptico

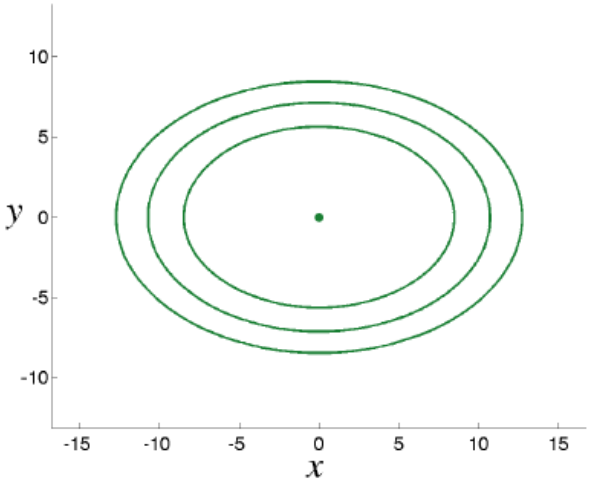




$$\begin{cases} z = \dfrac{k^2}{a^2} + \dfrac{y^2}{b^2} \\ x = k \end{cases} \quad : \quad \text{parábolas } \forall k \in \mathbb{R}.$$

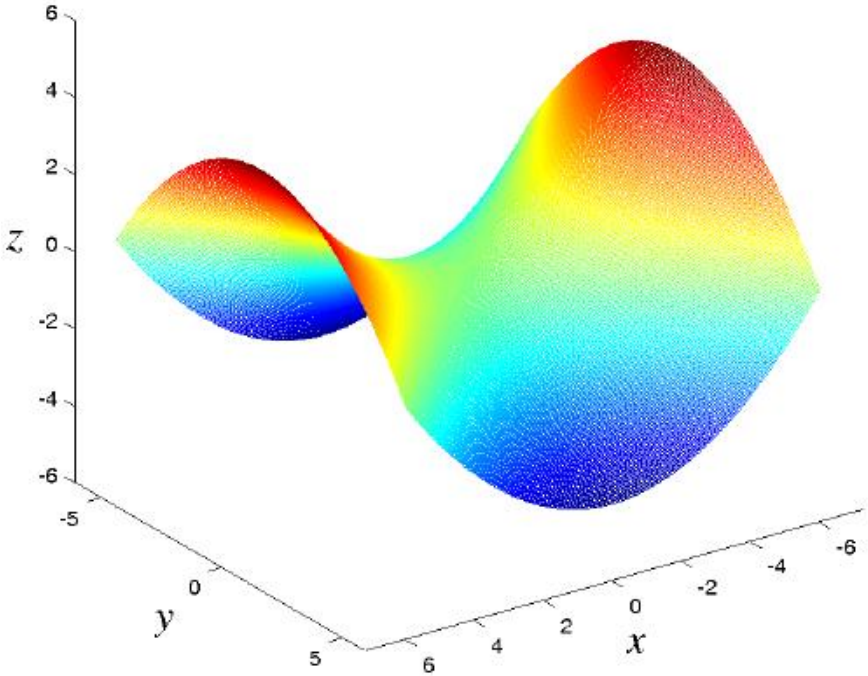
Parabolóide Elíptico





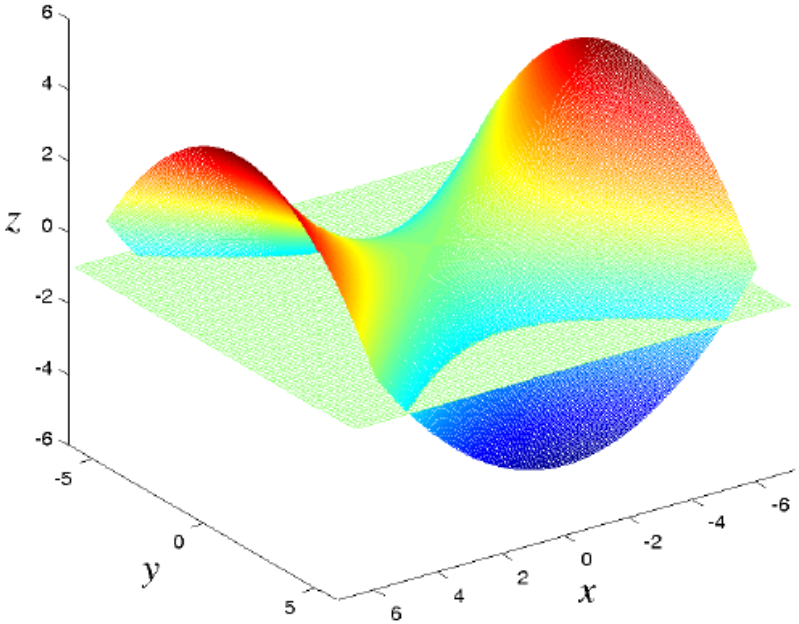
$$\begin{cases} \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = k \\ z = k \end{cases} \quad : \quad \text{elipses } (k > 0).$$

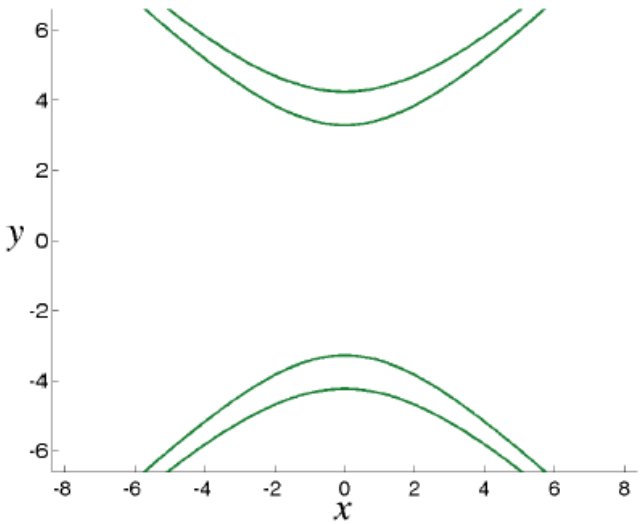
Parabolóide hiperbólico



$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

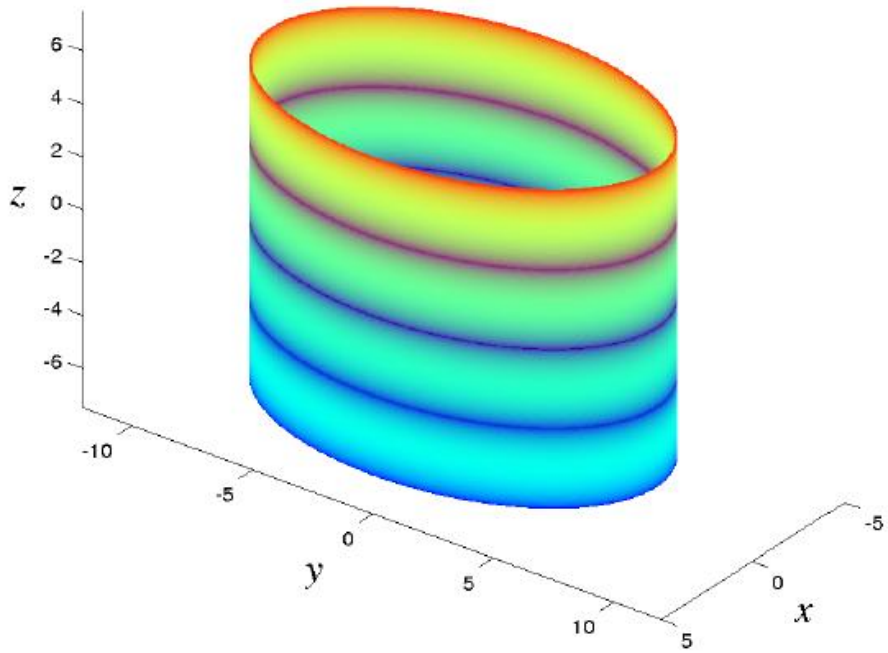
Parabolóide hiperbólico





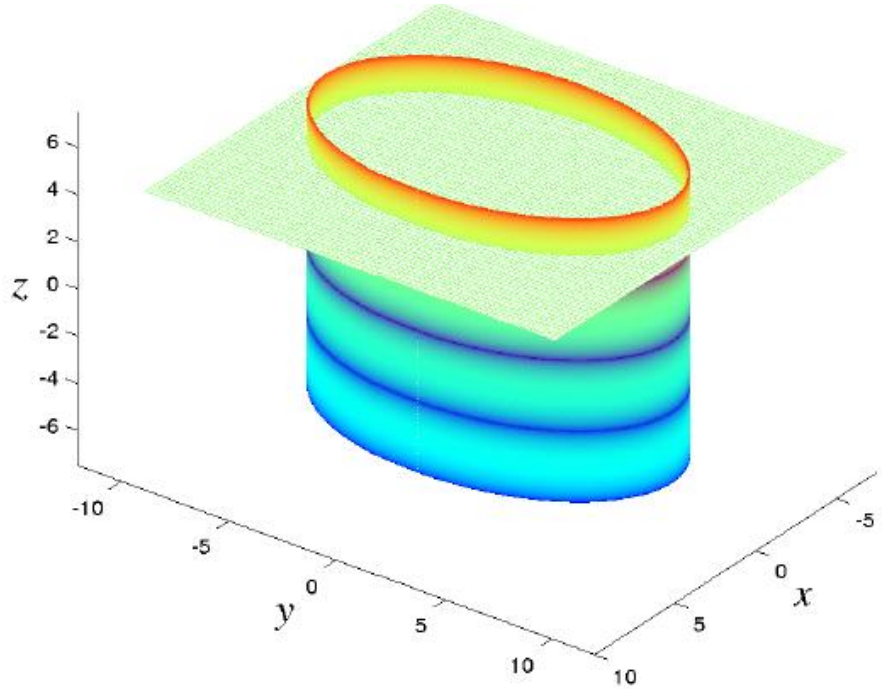
$$\begin{cases} -\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = -k \\ z = k \end{cases} \quad : \quad \text{hipérboles na vertical } \forall k < 0.$$

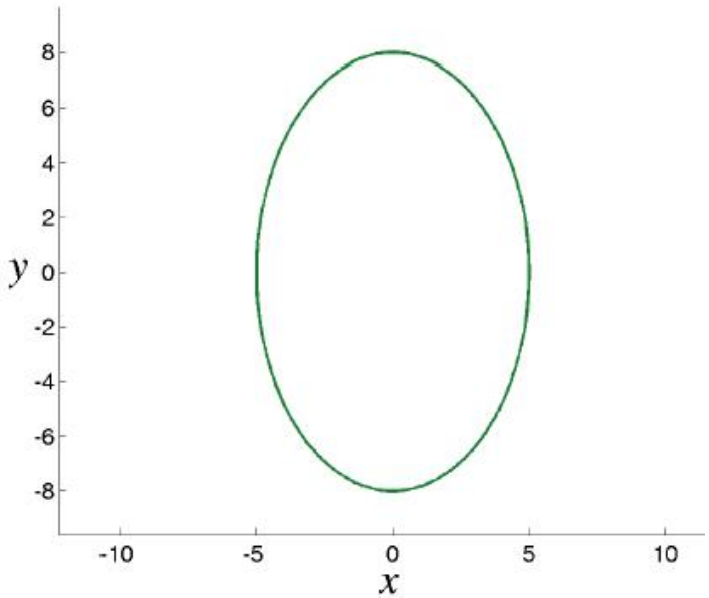
Cilindro elíptico



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

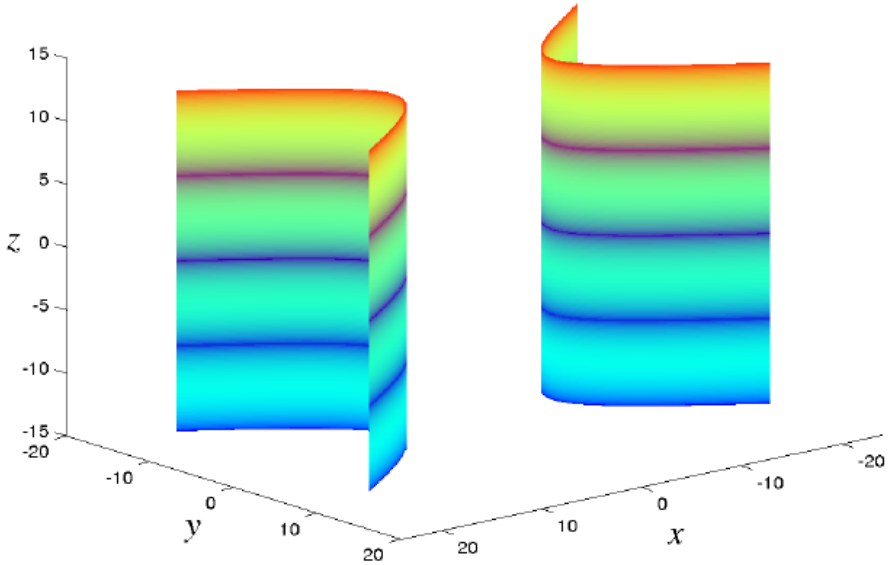
Cilindro elíptico





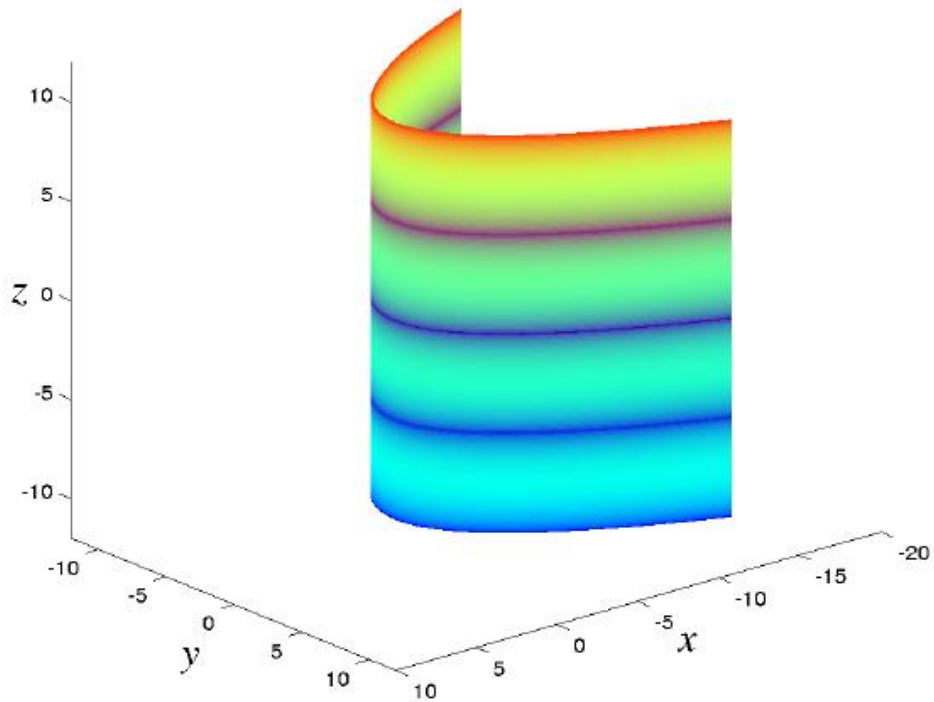
$$\begin{cases} \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \\ z = k \end{cases} \quad : \quad \text{elipses iguais } \forall k \in \mathbb{R}.$$

Cilindro hiperbólico



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Cilindro parabólico



$$y^2 = -2px$$

Fontes Bibliográficas:

http://www2.ufersa.edu.br/portal/view/uploads/setores/114/quadricas_conicas.pdf

http://www.eqm.unisul.br/disciplinas/ga/Unid4GeometriaEspaco.pdf

http://objetoseducacionais2.mec.gov.br/bitstream/handle/mec/10492/GeometriaAnaliticaQuadricas.pdf?sequence=1